

AI-Assisted Performance-Target-Driven Inverse Design for Energy-Efficient Office Facades

BPS5231 AI for Sustainable Building Design

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KeyWords

Performance-driven Design

Facade Design

Inverse Design

Building energy performance

Surrogate Modelling

Machine Modelling

Background

In traditional initial façade design, architects often use the forward design process. They first define the window dimensions and materials, then do energy simulations, and finally get performance results. However, architects often find that it is difficult to choose the right or suitable window-to-wall ratio at this very first stage. A certain window size may meet daylighting requirements, but it may also increase cooling energy use especially in the tropical regions. Traditional methods rely on repeated simulations and multiple comparisons. Essentially, this is a time-consuming process of trial and error, which makes it difficult to systematically explore the design space. As initiatives to improve energy efficiency and reduce carbon emissions in the construction sector gain momentum, monitoring and managing energy consumption on construction sites is becoming increasingly important from the earliest planning stages. Furthermore, the question of “If we have set a specific energy consumption target, what façade parameters should be used?” cannot be answered in the process of drafting regulations directly.

Research Gap + Research Question

Conventional Forward Design

Design Facade Parameters → Energy Simulation → EUI

Proposed Inverse Design

Target EUI → Surrogate Model+Optimization → Feasible Facade Design

The variables are WWR, SHGC, U-value, Shade Depth.

In the early stages of a project, how can a target building EUI be translated into feasible facade design parameter combinations?

Baseline Office Model Size

Baseline Office Plan: 35m*25m ¹

Area: 875m²

Floor to Floor: 4m ²

Number of floors: 15

Building Height: 60m

Gross Floor Area: 13125m²

Climate: Singapore (Changi EPW)

A simplified rectangular office geometry was used as a fixed reference model for controlled facade parametric studies. The building geometry remains fixed throughout the parametric study.

¹ UK office buildings archetypal model as methodological approach in development of regression models for predicting building energy consumption from heating and cooling demands

² Development of Whole-Building Energy Models for Detailed Energy Insights of a Large Office Building with Green Certification Rating in Singapore

Façade variables + Ranges

Proposed / Preliminary

Variables	Ranges
WWR	20%-80%
SHGC	0.2-0.7
U-value	1–4 W/m ² K
Shading depth	0–1.5 m

Data–Change WWR

Case	WWR	SHGC	U-value	ShadeDepth	EUI
1	0.2	0.4	2	0	123.897
2	0.4	0.4	2	0	133.183
3	0.6	0.4	2	0	141.311
4	0.8	0.4	2	0	147.478

Simulation engine: Honeybee + EnergyPlus

Climate: Singapore (Changi EPW)

GFA: 13,125 m²

Other parameters are held constant during the WWR sensitivity test.

Data–Change SHGC

Case	WWR	SHGC	U-value	ShadeDepth	EUI
1	0.4	0.2	2	0	124.499
2	0.4	0.4	2	0	133.183
3	0.4	0.6	2	0	141.728
4	0.4	0.7	2	0	145.698

Simulation engine: Honeybee + EnergyPlus

Climate: Singapore (Changi EPW)

GFA: 13,125 m²

Other parameters are held constant during the WWR sensitivity test.

Data–Change Shade Depth

Case	WWR	SHGC	U-value	ShadeDepth	EUI
1	0.4	0.4	2	0	133.183
2	0.4	0.4	2	0.5	129.29
3	0.4	0.4	2	1	127.049
4	0.4	0.4	2	1.5	125.923

Simulation engine: Honeybee + EnergyPlus

Climate: Singapore (Changi EPW)

GFA: 13,125 m²

Other parameters are held constant during the WWR sensitivity test.

Data–Change Shade U-value

Case	WWR	SHGC	U-value (W/m ² ·K)	ShadeDepth (m)	EUI (kWh/m ² ·yr)
1	0.4	0.4	1	0	132.016
2	0.4	0.4	2	0	133.183
3	0.4	0.4	3	0	133.131
4	0.4	0.4	4	0	134.117

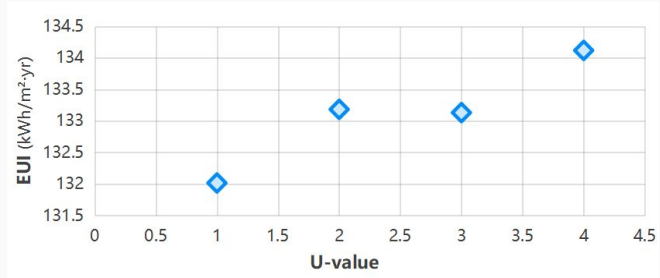
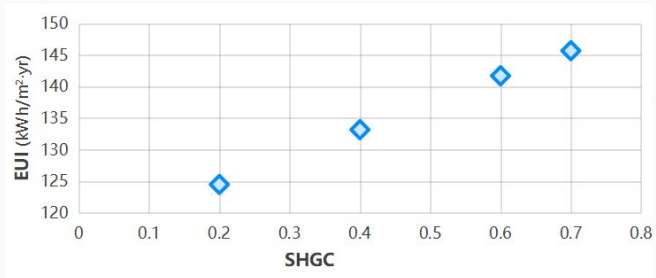
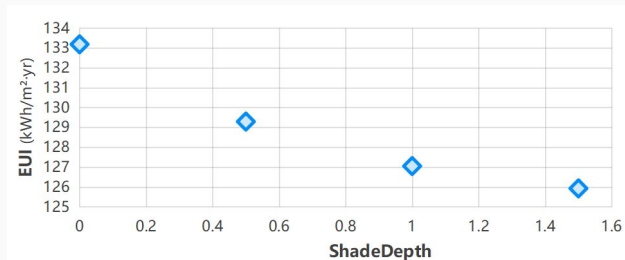
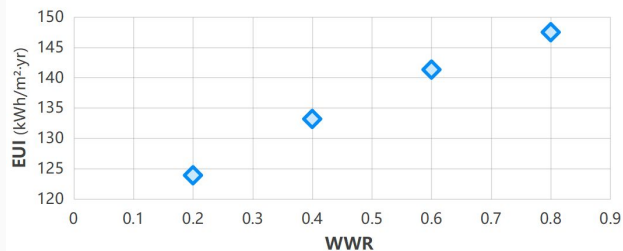
Simulation engine: Honeybee + EnergyPlus

Climate: Singapore (Changi EPW)

GFA: 13,125 m²

Other parameters are held constant during the WWR sensitivity test.

Preliminary EDA



Preliminary EDA

Preliminary Findings

- WWR and SHGC show strong positive relationships with EUI.
- Increasing shading depth reduces EUI, with diminishing marginal benefits.
- Window U-value shows a comparatively small influence within the tested range.

Potential Method

Phase 1– Parametric Modelling and Performance Data Generation

4 variables → Grasshopper + Honeybee + EnergyPlus → Datatest: $X \rightarrow EUI$

Phase 2–Surrogate Model Development

Training/validation set partitioning → Machine Learning Regression → Candidate models: Random Forest / XGBoost / Neural Network → Model Evaluation

Phase 3–Performance-Target-Driven Inverse Design

Input:target of EUI → Output:Multiple feasible facade parameter combinations

References/Related Works

1. Inverse design optimization of building EUI targets: generating diverse solutions and enhancing decision interpretability
2. Surrogate model-based design optimization of modular façades under multifunctional scenarios
3. A multi-stage optimization of passively designed high-rise residential buildings in multiple building operation scenarios
4. Optimizing the modular adaptive façade control strategy in open office space using integer programming and surrogate modelling